

STATE BOARD OF TECHNICAL EDUCATION AND TRAINING
TELANGANA
DIPLOMA EXAMINATION (C-21)
C21-END-NOV-24
SEMESTER III, SEMESTER END EXAM
AI/CCB/CPS/CS
CS-303/CPS-305
Computer Architecture



PCODE
13020

Exam Date: 14-11-2024
Duration: 2 Hours [11:30 AM To 01:30 PM]

Session: FN
[Total Marks: 40]

PART-A

Instructions:

1. Answer the following questions.
2. Each question carries **ONE** mark.

8 X 1 = 8

1. What is meant by Computer Architecture?
2. What is memory hierarchy in Computer?
3. List any 3 advantages of pipeline processing.
4. What is booth multiplication algorithm?
5. List any three Bus Systems.
6. List dis-advantage of Programmed I/O
7. What is Vector processing?
8. What is SIMD?

PART-B

Instructions:

1. Answer the following questions.
2. Each question carries **THREE** marks.

4 X 3 = 12

- 9(a). Convert Expression $X = (A+B) \times (C+D)$ into three address instruction format.
--- OR ---
- 9(b). Draw the block diagram of Input Output processor.
- 10(a). List any three differences between fixed point and floating point representation?
--- OR ---
- 10(b). Give the differences between CISC and RISC.
- 11(a). What is the need for an interface?
--- OR ---

- 11(b). Compare Synchronous and Asynchronous data transfer.
12(a). Write short notes on array processor.
----- OR -----
12(b). Write short notes on Instruction set architecture.

PART-C

Instructions:

1. Answer the following questions.
2. Each question carries FIVE marks.

4 X 5 = 20

- 13(a). Explain Direct address mode and Indirect address mode with an example.
----- OR -----
13(b). Explain Strobe controlled procedure of data transfer.
14(a). Explain the fixed point multiplication operation with flowchart.
----- OR -----
14(b). Explain about parallel processing.
15(a). What is bus system? explain various bus systems With the help of a neat diagram.
----- OR -----
15(b). Explain about daisy chaining priority with block diagram.
16(a). Explain instruction pipeline processing.
----- OR -----
16(b). Explain arithmetic pipeline with neat sketch.